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PROBLEM STATEMENT

01

Target Users:
System Analysts (SA) and Business Analysts (BA).

02

Current Practice: Manually reviewing meeting transcripts for hours to identify and extract requirements.

03

Problem: Time-consuming and cognitively demanding.

04

Risk: Prone to human error, potentially causing important information or agreements to be overlooked.

05

Impact: User Stories and Acceptance Criteria are often inconsistent and subjective.

WHY GENERATIVE AI?

01 **Meeting transcripts are conversational, unstructured, and often filled with small talk and linguistic variations.**

02 **Database-driven approaches, forms, and rule-based methods (Regex) struggle to capture semantic meaning.**

03 **LLMs act as semantic interpreters, filtering out noise while identifying relevant information.**

04 **Human-in-the-loop is essential: The LLM serves only as a drafting assistant and cannot be held accountable for the final requirements.**

System Architecture

01

Meeting transcripts are processed by the backend (Bun and PostgreSQL).

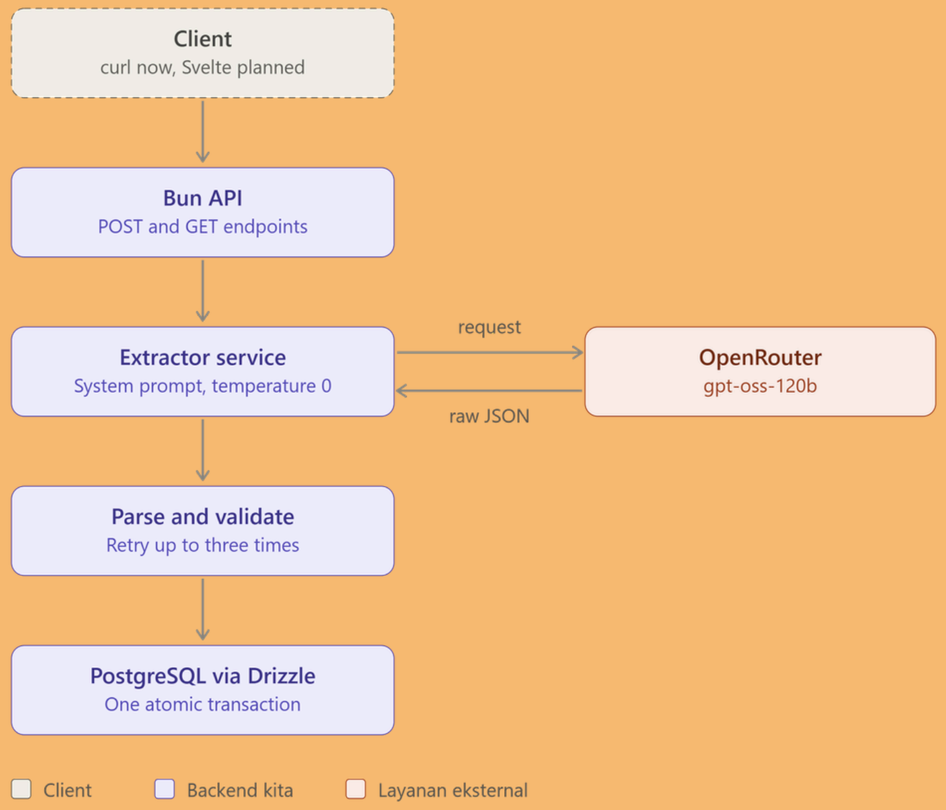




02

Extractor memanggil API OpenRouter (gpt-oss-120b) dengan suhu deterministik (0).





OUTPUT CONTROL

01

The JSON schema must include an out_of_scope array as an initial filtering mechanism.

02

The role, feature, and benefit fields must be free of introductory phrases.

03

Each User Story must represent only one role; combining multiple roles in a single story is prohibited.

04

Acceptance Criteria must strictly follow the Gherkin format (Given–When–Then).

05

Fault Tolerance: Markdown-cleaning regex, JSON isolation, and a maximum of three retry attempts.

TOOL

01

Current Limitation: OpenRouter API calls do not yet meet the criteria for Tool Calling.

02

Current Status: The system currently uses a standard model gateway invocation.

03

Planned Tool 1: Backlog Issue Exporter (integration with Jira/GitLab)

04

Planned Tool 2: Project Glossary & Domain Term Checker.

05

Authorization Requirement: The export tool must pass through a human authorization gate before execution.

GROUNDING & RAG

01.

Current Limitation: The grounding and Retrieval-Augmented Generation (RAG) layers have not yet been implemented.

02.

Current Risk: The system is vulnerable to hallucinations because it relies solely on in-context prompting.

03.

Planned RAG Implementation: Use five reference documents, including the INVEST guidelines, Software Requirements Specification (SRS), and other relevant standards.

04.

Chunking Strategy: Split documents along chapter boundaries, with 10–15% overlap between chunks.

05.

Hallucination Control: The AI must return “NOT FOUND IN REFERENCES” when no supporting evidence is available in the reference documents.

PRELIMINARY EVALUATION RESULTS

01

Evaluation Method: Five consecutive runs per test case with the temperature set to 0.

02

Success — Clinic Case: 0% leakage of prohibited feature keywords across all five runs.

03

Failure — Library Case: 60% failure rate, with prohibited feature keywords leaking in three out of five runs.

04

Failure Details: The words “reservation” and “booking” still appeared in Gherkin clauses.

05

Conclusion: Negative prompting alone is insufficient to reliably constrain LLM outputs.

RISKS AND ACTION PLAN

01

Key Risks: UI detail hallucinations and non-deterministic output variations.

02

Week 9: Implement RAG using five reference documents.

03

Weeks 10–13: Develop the Exporter and Validator tools with a human-in-the-loop mechanism.

04

Weeks 14–16: Test 14 test cases and evaluate latency, cost metrics, and ethical considerations.